

Name:

BUID:

Workshop 5: Quadratic Forms

1. (Review) Determine the least-squares best-fit line for the data points $\{(0,3), (1,1), (-1,1), (2,3)\}$ using the least-squares method

2. (Review) Determine the design matrix for finding the best-fit cubic polynomial for the data points in the previous problem

3. Determine an orthogonal diagonalization of $\begin{bmatrix} 1 & 3 \\ 3 & 1 \end{bmatrix}$

4. Expand the quadratic form $Q(x_1, x_2) = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 3 & -2 \\ -2 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ into a multivariate polynomial

5. Expand the quadratic form $Q(\mathbf{x}) = \mathbf{x}^T \begin{bmatrix} 1 & 2 & -1 \\ 2 & 3 & 0 \\ -1 & 0 & 5 \end{bmatrix} \mathbf{x}$ into a multivariate polynomial

6. Determine the symmetric matrix A such that $Q(x_1, x_2, x_3) = 5x_1^2 + 3x_2^2 + 2x_3^2 - x_1x_2 + 8x_2x_3$ can be written as $Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$

7. Determine if $Q(x_1, x_2, x_3) = 3x_1^2 + x_2^2 + 4x_2x_3 + x_3^2$ is positive definite, positive semidefinite, indefinite, or negative definite

~~8. Determine $\max_{x_1^2+x_2^2=1} 3x_1^2+7x_2^2$ and $\min_{x_1^2+x_2^2=1} 3x_1^2+7x_2^2$~~

8/9. Let $Q(\mathbf{x})$ be the quadratic form from problem 7. Determine $\max_{\|\mathbf{x}\|=1} Q(\mathbf{x})$ and $\min_{\|\mathbf{x}\|=1} Q(\mathbf{x})$

9/10. Let $Q(\mathbf{x})$ be the quadratic form from problem 7. Determine an orthogonal diagonalization of its underlying matrix

10/11. Give an example of a negative definite quadratic form over two variables that has all positive entries

Write your solutions below and on the back on this page.

$$1. \quad X = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & -1 \\ 1 & 2 \end{bmatrix} \quad \vec{y} = \begin{bmatrix} 3 \\ 1 \\ 1 \\ 3 \end{bmatrix} \quad X^T X = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & -1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & -1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 4 & 2 \\ 2 & 6 \end{bmatrix}$$

$$X^T \vec{y} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & -1 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \\ 1 \\ 3 \end{bmatrix} = \begin{bmatrix} 8 \\ 6 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 2 & 8 \\ 2 & 6 & 6 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & 3 \\ 2 & 1 & 4 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & 3 \\ 0 & -5 & -2 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & 3 \\ 0 & 1 & 2/5 \end{bmatrix}$$

$$f(x) = 9/5 + 2/5 x$$

$$\sim \begin{bmatrix} 1 & 0 & 9/5 \\ 0 & 1 & 2/5 \end{bmatrix}$$

Continue your work here.

2. $f(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3$

$$X = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 1 & 2 & 4 & 8 \end{bmatrix}$$

$$y = \begin{bmatrix} 3 \\ 1 \\ 1 \\ 3 \end{bmatrix}$$

3. $\det(A - \lambda I) = (\lambda - 1)^2 - 9 = \lambda^2 - 2\lambda - 8 = (\lambda - 4)(\lambda + 2) \quad \lambda = 4, -2$

$$A - 4I = \begin{bmatrix} -3 & 3 \\ 3 & -3 \end{bmatrix} \sim \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix} \quad \begin{matrix} x_1 = x_2 \\ x_2 \text{ is free} \end{matrix} \quad \vec{v} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \frac{\vec{v}}{\|\vec{v}\|} = \begin{bmatrix} r_{1/2} \\ r_{1/2} \end{bmatrix}$$

$$A + 2I = \begin{bmatrix} 3 & 3 \\ 3 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \quad \begin{matrix} x_1 = -x_2 \\ x_2 \text{ is free} \end{matrix} \quad \vec{v} = \begin{bmatrix} -1 \\ 1 \end{bmatrix} \quad \frac{\vec{v}}{\|\vec{v}\|} = \begin{bmatrix} -r_{1/2} \\ r_{1/2} \end{bmatrix}$$

$$A = \underbrace{\begin{bmatrix} r_{1/2} & -r_{1/2} \\ r_{1/2} & r_{1/2} \end{bmatrix}}_P \underbrace{\begin{bmatrix} 4 & 0 \\ 0 & -2 \end{bmatrix}}_D \underbrace{\begin{bmatrix} r_{1/2} & r_{1/2} \\ -r_{1/2} & r_{1/2} \end{bmatrix}}_{P^{-1} = P^T}$$

4. $Q(x_1, x_2) = 3x_1^2 - 4x_1x_2 + 7x_2^2$

5. $Q(x_1, x_2, x_3) = x_1^2 + 3x_2^2 + 5x_3^2 + 4x_1x_2 - 2x_1x_3$

6. $Q(\vec{x}) = \vec{x}^T \begin{bmatrix} 5 & -1/2 & 0 \\ -1/2 & 3 & 4 \\ 0 & 4 & 2 \end{bmatrix} \vec{x}$

7. $A = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 1 & 2 \\ 0 & 2 & 1 \end{bmatrix} \quad \det(A - \lambda I) = (3 - \lambda)((\lambda - 1)^2 - 4) = (3 - \lambda)(\lambda^2 - 2\lambda - 3) = (3 - \lambda)(\lambda - 3)(\lambda + 1) = -(\lambda - 3)^2(\lambda + 1) \quad \lambda = 3, -1$

Indefinite

8. $\max_{\|\vec{x}\|=1} Q(\vec{x}) = \max_{\|\vec{x}\|=1} \vec{x}^T A \vec{x} = 3 \quad \min_{\|\vec{x}\|=1} Q(\vec{x}) = -1$

9. $A - 3I = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -2 & 2 \\ 0 & 2 & -2 \end{bmatrix} \sim \begin{bmatrix} 0 & 1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{matrix} x_1 \text{ is free} \\ x_2 = x_3 \\ x_3 \text{ is free} \end{matrix} \quad \vec{v}_1 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \quad \frac{\vec{v}_1}{\|\vec{v}_1\|} = \begin{bmatrix} 0 \\ r_{1/2} \\ r_{1/2} \end{bmatrix} \quad \vec{v}_2 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

$$A + I = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 2 & 2 \\ 0 & 2 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \quad \begin{matrix} x_1 = 0 \\ x_1 = -x_2 \\ x_2 \text{ is free} \end{matrix} \quad \vec{v} = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} \quad \frac{\vec{v}}{\|\vec{v}\|} = \begin{bmatrix} 0 \\ -r_{1/2} \\ r_{1/2} \end{bmatrix}$$

10. $A = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & r_{1/2} & -r_{1/2} \\ 0 & r_{1/2} & r_{1/2} \end{bmatrix}}_P \underbrace{\begin{bmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & -1 \end{bmatrix}}_D \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & r_{1/2} & r_{1/2} \\ 0 & -r_{1/2} & r_{1/2} \end{bmatrix}}_{P^{-1} = P^T}$